

The 2D SYK Vacuum: IR Structure, JT Duality, and the Architecture of the WIN Paradigm

Stanley Preschutti

Entropia Research Institute / Information Physics Institute

ORCID: 0009-0004-5445-1744

September 2026

Abstract

We compute the large- N infrared structure of the SYK array defined on the 8×8 periodic torus with one Majorana fermion per site ($N = 64$). For the standard uncorrelated disorder ensemble, we prove that the melon-equation self energy is exactly diagonal in position and therefore momentum independent. The infrared propagator is k -independent to leading order, and the theory flows to the 0+1-dimensional SYK conformal fixed point with scaling dimension $\Delta = 1/4$. The holographic dual is AdS_2 , not AdS_4 . We extend the analysis numerically to momentum-conserving disorder at $N = 16$ and find that the position-independence persists to 0.64%. We conclude that the two-dimensional spatial structure is infrared-irrelevant and that the SYK array does not produce a four-dimensional gravitational dual. We then state the corrected architecture of the WIN Paradigm, in which the $D = 4$ substrate is provided by the FDLM/WIN layer and the SYK/QIN layer serves as the microscopic population and stabilization mechanism. Using this corrected architecture, we catalogue six errors that appear in previously circulated WIN papers and provide the corrected statements.

Contents

1	Introduction	3
2	The SYK Array on the Torus	3
2.1	Lattice and fermions	3
2.2	Hamiltonian	3
2.3	Disorder ensembles	4
2.4	Green's functions	4
3	Position-Independence of the Self-Energy	4
3.1	The melon self-energy	4
3.2	Theorem	4
3.3	Momentum-space consequence	5
3.4	Momentum-conserving ensemble	5
4	The Infrared Fixed Point	5
4.1	Dyson equation	5
4.2	Infrared limit	5
4.3	Self-consistency	6
4.4	Conformal solution	6
4.5	Holographic dual	6

5	Numerical Test at $N = 16$ with Momentum-Conserving Disorder	6
5.1	Setup	6
5.2	Result	7
5.3	Interpretation	7
5.4	Finite-size caveat	7
6	What Would Be Required for Two-Dimensional Structure to Survive	7
6.1	Condition on the disorder kernel	7
6.2	Physical interpretation	7
6.3	Relevance to the WIN Paradigm	8
7	The Corrected Architecture of the WIN Paradigm	8
7.1	Statement	8
7.2	What SYK does and does not do	8
7.3	Why this is the correct architecture	9
8	Corrections to Previous Papers	9
8.1	Correction 1 — The value of γ	9
8.2	Correction 2 - The Gaussian variance	9
8.3	Correction 3 - The relation between γ and b	9
8.4	Correction 4 - The chaos gravity correspondence	9
8.5	Correction 5 - 4D gravity from 2D SYK	10
8.6	Correction 6 - The role of SYK in the overall architecture	10
9	Conclusion	10
A	Numerical Values	11
B	Reproducible Verification	11
C	Summary of Corrections	13

1 Introduction

The WIN Paradigm is a six layer framework in which a discrete substrate, a dimensional stability theorem, a dynamical stability sector, an effective field theory of the vacuum, a master selection framework, and a microscopic model are arranged in a dependency chain. The microscopic layer is a Sachdev-Ye-Kitaev (SYK) model [1, 2] on the 8×8 torus, consisting of one Majorana fermion per site with all-to-all random quartic couplings. This layer has been described in previous WIN papers as producing emergent four-dimensional gravity, as giving rise to a chaos-gravity correspondence $G_{\text{eff}} = 1/\lambda_L$, and as being UV finite through a Gaussian regulator with stability coefficient $\gamma \approx 1.0\text{--}1.2$.

This paper establishes that the first two of these claims are incorrect and that the third is based on an arithmetic error. The SYK array on the torus does not produce four-dimensional gravity. Its infrared fixed point is the 0+1-dimensional SYK conformal field theory, dual to Jackiw-Teitelboim (JT) gravity in two dimensions, not to Einstein gravity in four. The chaos-gravity correspondence, understood as a quantitative relation between the Newton constant and the Lyapunov exponent, does not hold in any temperature regime; the two quantities are both set by the Schwarzian coupling but are not inverse functions of one another. The Gaussian regulator coefficient $\gamma \approx 1.0\text{--}1.2$ was computed from a variance that was off by a factor of two and a relation $\gamma = \sqrt{2|b|}$ that has no derivation; the corrected coefficient is $\gamma = -b \approx 0.62$, consistent with the DME stability coefficient $\gamma = 0.702839\dots$ at the level of the numerical uncertainty in b , though the two γ 's describe different physical scales.

The positive content of this paper is a theorem on the large- N infrared structure of the SYK array. For the standard uncorrelated disorder ensemble, the melon-equation self-energy is exactly position diagonal. The Fourier-transformed self energy is momentum-independent. The Dyson equation then reduces, in the infrared, to the standard 0+1-dimensional SYK self consistency equation for the local Green's function, and the theory flows to the 0+1-dimensional conformal fixed point. We verify this numerically at $N = 16$ for momentum conserving disorder, finding position-independence to sub-percent precision.

The paper is organized as follows. Section 2 sets up the model. Section 3 establishes the position-independence theorem. Section 4 derives the infrared fixed point. Section 5 presents the numerical test. Section 6 discusses what would be required for two-dimensional structure to survive. Section 7 states the corrected architecture of the WIN Paradigm. Section 8 catalogues the corrections to previous papers. Section 9 concludes.

2 The SYK Array on the Torus

2.1 Lattice and fermions

Let $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ denote the periodic square lattice with $N = 64$ sites. On each site $x \in \Lambda$ place one Majorana fermion ψ_x satisfying the anticommutation relations

$$\{\psi_x, \psi_{x'}\} = 2\delta_{xx'}. \quad (1)$$

2.2 Hamiltonian

The Hamiltonian is

$$H = \sum_{x_1, x_2, x_3, x_4} J(x_1, x_2, x_3, x_4) \psi_{x_1} \psi_{x_2} \psi_{x_3} \psi_{x_4}, \quad (2)$$

where the sum runs over all ordered quadruples of sites and the couplings J are real. Hermiticity of H requires J to be antisymmetric under exchange of any pair of indices.

2.3 Disorder ensembles

We consider two disorder ensembles.

Uncorrelated ensemble. The couplings $J(x_1, x_2, x_3, x_4)$ are independent Gaussian random variables with zero mean and variance

$$\overline{J(x_1, x_2, x_3, x_4)^2} = \frac{J^2}{N^3}. \quad (3)$$

Here and throughout, an overline denotes averaging over the disorder ensemble.

Momentum-conserving ensemble. The couplings satisfy the additional constraint

$$J(x_1, x_2, x_3, x_4) = 0 \quad \text{unless} \quad k_{x_1} + k_{x_2} + k_{x_3} + k_{x_4} \equiv 0 \pmod{8}, \quad (4)$$

where $k_x = (n_1(x), n_2(x))$ is the momentum label of site x . For nonvanishing couplings, the variance is the same as in the uncorrelated case.

2.4 Green's functions

The Euclidean Green's function is

$$G(x, x'; \tau) = \langle T_\tau \psi_x(\tau) \psi_{x'}(0) \rangle, \quad (5)$$

where T_τ denotes time-ordering and the expectation is in the thermal ensemble at inverse temperature β . The Fourier transform in time is denoted $G(x, x'; i\omega)$. The local Green's function is

$$G_{\text{loc}}(i\omega) = \frac{1}{N} \sum_x G(x, x; i\omega). \quad (6)$$

3 Position-Independence of the Self-Energy

3.1 The melon self-energy

At leading order in $1/N$, the self-energy is the sum of melon diagrams:

$$\Sigma(x, x'; \tau) = \sum_{y_1, y_2, y_3} \sum_{z_1, z_2, z_3} \overline{J(x, y_1, y_2, y_3) J(x', z_1, z_2, z_3)} G(y_1, z_1; \tau) G(y_2, z_2; \tau) G(y_3, z_3; \tau). \quad (7)$$

3.2 Theorem

Theorem 1 (Position-independence). *For the uncorrelated disorder ensemble, the large- N self-energy satisfies*

$$\Sigma(x, x'; \tau) = \delta_{xx'} \Sigma_0(\tau), \quad (8)$$

where

$$\Sigma_0(\tau) = \frac{J^2}{N^3} \sum_{y_1, y_2, y_3} G(y_1, y_1; \tau) G(y_2, y_2; \tau) G(y_3, y_3; \tau). \quad (9)$$

Proof. For the uncorrelated ensemble,

$$\overline{J(x, y_1, y_2, y_3) J(x', z_1, z_2, z_3)} = \frac{J^2}{N^3} \delta_{xx'} \delta_{y_1 z_1} \delta_{y_2 z_2} \delta_{y_3 z_3} \quad (10)$$

after symmetrization over the three internal pairs. Substituting into the melon expression yields

$$\Sigma(x, x'; \tau) = \delta_{xx'} \frac{J^2}{N^3} \sum_{y_1, y_2, y_3} G(y_1, y_1; \tau) G(y_2, y_2; \tau) G(y_3, y_3; \tau), \quad (11)$$

which is the stated result. $\square$

3.3 Momentum-space consequence

Define the Fourier transform

$$\Sigma(k, i\omega) = \sum_x e^{-ik \cdot x} \Sigma(x, 0; i\omega). \quad (12)$$

By Theorem 1,

$$\Sigma(k, i\omega) = \Sigma_0(i\omega), \quad (13)$$

independent of k .

This is the central structural result of the paper. In the uncorrelated SYK array, the self energy does not know about the position of the external fermion. It has no momentum dependence at leading order in $1/N$.

3.4 Momentum-conserving ensemble

For the momentum-conserving ensemble, the disorder variance acquires the constraint

$$\overline{J(x_1, x_2, x_3, x_4) J(x'_1, x'_2, x'_3, x'_4)} = \frac{J^2}{N^3} \prod_{i=1}^4 \delta_{x_i x'_i} \Theta(k_{x_1} + k_{x_2} + k_{x_3} + k_{x_4}), \quad (14)$$

where Θ enforces momentum conservation. The self-energy becomes

$$\begin{aligned} \Sigma(x, x'; \tau) &= \frac{J^2}{N^3} \sum_{y_1, y_2, y_3} \Theta(k_x + k_{y_1} + k_{y_2} + k_{y_3}) \Theta(k_{x'} + k_{y_1} + k_{y_2} + k_{y_3}) \\ &\quad \times G(y_1, y_1; \tau) G(y_2, y_2; \tau) G(y_3, y_3; \tau). \end{aligned} \quad (15)$$

The two Θ constraints force $k_x = k_{x'}$, so the self-energy is again diagonal in position. The momentum-conserving constraint does not spoil the position-independence; it merely restricts the sum over intermediate momenta. The k -dependence of the momentum-space self-energy is $O(1/N)$, vanishing in the large- N limit.

We verify this numerically in Section 5.

4 The Infrared Fixed Point

4.1 Dyson equation

The Fourier-space Dyson equation is

$$G(k, i\omega)^{-1} = i\omega - \epsilon_k - \Sigma_0(i\omega), \quad (16)$$

where $\epsilon_k = 4 - 2 \cos(2\pi k_1/8) - 2 \cos(2\pi k_2/8)$ is the torus dispersion. The self-energy Σ_0 is independent of k , so the k -dependence of the propagator is entirely in the bare dispersion.

4.2 Infrared limit

Assume the standard SYK infrared condition,

$$|\Sigma_0(i\omega)| \gg \max(|\omega|, |\epsilon_k|), \quad (17)$$

for $\omega \rightarrow 0$. Then

$$G(k, i\omega) \approx -\frac{1}{\Sigma_0(i\omega)} \left[1 + \frac{\epsilon_k - i\omega}{\Sigma_0(i\omega)} + O\left(\frac{\omega^2}{\Sigma_0^2}\right) \right]. \quad (18)$$

The leading term is independent of k . The subleading correction is suppressed by $1/|\Sigma_0| \sim |\omega|^{3/2}$.

4.3 Self-consistency

To leading order in the infrared, define the local Green's function

$$G_{\text{loc}}(i\omega) = -\frac{1}{\Sigma_0(i\omega)}. \quad (19)$$

The self-consistency equation becomes

$$\Sigma_0(\tau) = J^2 G_{\text{loc}}(\tau)^3, \quad (20)$$

which is exactly the 0+1-dimensional SYK self-consistency equation.

4.4 Conformal solution

The 0+1-dimensional SYK equation has the conformal infrared solution

$$G_{\text{loc}}(\tau) \sim \frac{1}{|\tau|^{2\Delta}}, \quad \Sigma_0(\tau) \sim \frac{1}{|\tau|^{2(1-\Delta)}}, \quad (21)$$

with scaling dimension

$$\boxed{\Delta = \frac{1}{4}}. \quad (22)$$

The exponent is fixed by the self-consistency equation $G \sim \Sigma^{-1}$ together with $\Sigma \sim G^3$: if $G \sim \tau^{-2\Delta}$, then $\Sigma \sim \tau^{-6\Delta}$, and $\Sigma \sim G^{-1}$ requires $6\Delta = 2\Delta$, giving $2\Delta = 1/2$ and $\Delta = 1/4$.

4.5 Holographic dual

The 0+1-dimensional SYK conformal field theory at low temperature is dual to Jackiw-Teitelboim gravity in two dimensions [3, 4]. The bulk is nearly AdS_2 . The Schwarzian mode at the boundary is the hydrodynamic variable of the dual gravitational theory.

The bulk does not have a propagating graviton. JT gravity is a topological theory of the two dimensional metric; the only degree of freedom is the boundary Schwarzian mode. There is no massless spin-2 particle in the spectrum.

5 Numerical Test at $N = 16$ with Momentum-Conserving Disorder

5.1 Setup

We consider a 4×4 torus with $N = 16$ Majorana fermions. The Hamiltonian

$$H = \sum_{i < j < k < l} J_{ijkl} \psi_i \psi_j \psi_k \psi_l \quad (23)$$

is constructed with momentum-conserving Gaussian couplings: $J_{ijkl} \neq 0$ only when $k_i + k_j + k_k + k_l \equiv 0 \pmod{4}$, with $J_{ijkl} \sim \mathcal{N}(0, 6J^2/N^3)$. The Hilbert space dimension is $2^8 = 256$, tractable by exact diagonalization.

We compute the position-diagonal Green's function

$$G(x, x; \tau) = \langle T_\tau \psi_x(\tau) \psi_x(0) \rangle \quad (24)$$

at inverse temperature $\beta = 2.0$ and $\tau = \beta/2$, using the Lehmann representation in the eigenbasis of H .

5.2 Result

The per-site values are:

$$\mu_G = 0.9531, \quad \sigma_G = 0.0061, \quad \min = 0.9424, \quad \max = 0.9637. \quad (25)$$

The fractional variation is

$$\boxed{\frac{\sigma_G}{\mu_G} = 0.0064 = 0.64\%}. \quad (26)$$

5.3 Interpretation

The Green's function is position independent to sub-percent precision. The 2D spatial structure is not preserved in the infrared. The theory flows to 0+1-dimensional SYK.

The momentum conserving constraint is too weak to preserve the 2D structure. The reason is that the constraint removes $O(1)$ modes out of $O(N)$ in the momentum sum; it therefore affects the self-energy only at $O(1/N)$, which vanishes in the large- N limit. This is consistent with the analytical result of Section 3.

5.4 Finite-size caveat

The test is at $N = 16$, not $N = 64$. A larger system would give a smaller fractional variation, but the structural argument of Section 3 is independent of N . We expect the 0.64% variation to decrease as N increases, consistent with the $O(1/N)$ scaling of the momentum-dependent corrections.

6 What Would Be Required for Two-Dimensional Structure to Survive

6.1 Condition on the disorder kernel

For the self energy to acquire momentum dependence at leading order in $1/N$, the disorder variance must be non-trivial in the position pairings. The general form is

$$\overline{J(x_1, x_2, x_3, x_4)^2} = \frac{J^2}{N^3} V(x_1 - x_3) V(x_2 - x_4), \quad (27)$$

where V is a translation invariant kernel on the torus. Fourier-transforming gives

$$\Sigma(k, i\omega) - \Sigma_0(i\omega) = \frac{J^2}{N^3} \hat{V}(k) \tilde{G}(k, i\omega), \quad (28)$$

where $\tilde{G}$ is a convolution of three Green's functions.

For the correction to survive in the infrared, we require $\hat{V}(0) \neq 0$, i.e., the kernel has a nonzero zero-momentum component. If $\hat{V}(0) = 0$, the kernel is “purely exchange” and the leading correction vanishes.

6.2 Physical interpretation

A kernel with $\hat{V}(0) = 0$ corresponds to a four-fermion interaction that has no contact term. Such interactions are non-generic. They require the four-fermion coupling to vanish for zero-momentum external states.

6.3 Relevance to the WIN Paradigm

Whether such a kernel can arise naturally in the WIN Paradigm is an open question. However, it is not required for the WIN architecture, because the SYK layer does not need to produce the four-dimensional substrate. That is the role of the FDLM/WIN layer, which derives $D = 4$ from the photon's two helicity states and the resulting four-dimensional phase space.

7 The Corrected Architecture of the WIN Paradigm

7.1 Statement

The WIN Paradigm consists of six layers. Their roles are:

1. **FDLM/WIN.** Provides the discrete substrate. The 8×8 torus with $N = 64$ sites, the 14 modes at $\lambda = 4$, the support classes, the zero state, the mass map, and the baseline $w = -1$. The $D = 4$ dimension is derived from the photon's two helicity states. This layer is independent of SYK.
2. **DME.** Provides the dimensional stability coefficient $\gamma = 0.702839467990 \dots$. The entropy well $S/S_0 = 1 - \gamma\epsilon^2 + O(\epsilon^4)$ stabilizes $D = 4$.
3. **ENTROPIA-Core.** Provides the dynamical stability sector: the Lyapunov functional, the gradient flow, and the master conditions M1 and M2.
4. **ENTROPIA-EFT.** Provides the effective field theory of the vacuum: the equation of state, the bulk viscosity, the Proca sector, and the gravitational coupling.
5. **ENTROPIX/MESA.** Provides the master selection framework.
6. **SYK/QIN.** Provides the microscopic population and stabilization mechanism. The SYK array on the torus defines the fermionic degrees of freedom, satisfies the antipode operator identity $\mathcal{O}_{A(m)} = -\mathcal{O}_m$, provides the maximal chaos that saturates the Maldacena-Shenker-Stanford bound, and supplies the JT duality in two dimensions.

7.2 What SYK does and does not do

Does:

- Defines the microscopic fermionic degrees of freedom.
- Satisfies the antipode operator identity, which stabilizes the mode structure.
- Provides maximal chaos, saturating the MSS bound.
- Supplies the JT dual in two dimensions.

Does not:

- Produce the 4D geometry. That is FDLM/WIN's role.
- Produce a propagating graviton.
- Give a massless spin-2 pole in the boundary stress-tensor correlator.
- Require momentum-dependent structure in the infrared.

7.3 Why this is the correct architecture

The SYK array is a 0+1-dimensional quantum mechanical system in the infrared, regardless of the two-dimensional spatial structure. Its JT dual is AdS_2 . The four-dimensional bulk that appears in the WIN Paradigm is provided by the FDLM/WIN substrate, not by SYK.

Attempting to derive 4D gravity from a 2D SYK array is a category error: SYK is dual to JT gravity, which is 2D, not to Einstein gravity, which is 4D.

8 Corrections to Previous Papers

We catalogue six corrections that should be applied to previously circulated WIN papers.

8.1 Correction 1 — The value of γ

Previous statement. SYK TENSOR papers assert $\gamma \approx 1.0$ – 1.2 .

Corrected statement. The stability coefficient is

$$\gamma = 0.70283946799007245929257819650854315\dots, \quad (29)$$

as derived from the DME strip integral.

Source of error. The SYK TENSOR value was a coefficient of a different ϵ^2 (the Gaussian regulator width) and not the DME coefficient (the dimensional deviation). The two coefficients describe unrelated scales and are not interchangeable.

8.2 Correction 2 - The Gaussian variance

Previous statement. UV and soft-regulation papers compute $\sigma^2 = 1/4$.

Corrected statement. The Gaussian variance is

$$\sigma^2 = 1/2. \quad (30)$$

Source of error. The ratio $\Gamma(3/2)/\Gamma(1/2)$ equals $1/2$, not $1/4$. The earlier calculation inserted an additional factor of $1/2$.

8.3 Correction 3 - The relation between γ and b

Previous statement. UV papers assert $\gamma = \sqrt{2|b|}$ in the Markovian limit.

Corrected statement. If γ and b describe the same entropy with the same normalization, the correct relation is

$$\gamma = -b. \quad (31)$$

With $|b| \approx 0.62$, this gives $\gamma \approx 0.62$, consistent with DME's value at the level of the numerical uncertainty.

Source of error. The relation $\gamma = \sqrt{2|b|}$ has no derivation. It appears to conflate a Gaussian variance relation with a coefficient identity.

8.4 Correction 4 - The chaos gravity correspondence

Previous statement. SYK TENSOR papers report a numerical anticorrelation $r = -0.94$ between G_{eff} and λ_L , and interpret this as evidence for $G_{\text{eff}} = 1/\lambda_L$.

Corrected statement. In JT gravity, the Newton constant and the Lyapunov exponent are both fixed by the Schwarzian coupling $C = N/(4J)$, but they scale differently. At low temperature,

$$G_N \lambda_L \sim \frac{JT}{4N}, \quad (32)$$

temperature-dependent. At high temperature,

$$G_N \lambda_L \sim \frac{cJ^2}{8\pi N}, \quad (33)$$

coupling-dependent. Neither regime gives $G_N \lambda_L = \text{constant}$. The reported correlation is either tautological (if G_{eff} is defined as $1/\lambda_L$) or a finite- N artifact.

Corrected relation. G_N and λ_L are both determined by the same Schwarzian data, but they are not inverses of one another. The chaos-gravity correspondence is qualitative, not quantitative.

8.5 Correction 5 - 4D gravity from 2D SYK

Previous statement. SYK TENSOR papers claim that four-dimensional gravity emerges from the scrambling dynamics of the SYK array.

Corrected statement. The 2D SYK array flows to 0+1-dimensional SYK in the infrared. Its holographic dual is AdS_2 , not AdS_4 . There is no propagating graviton. The correct statement is that SYK is dual to JT gravity in two dimensions.

Corrected architecture. The four-dimensional substrate is provided by FDLM/WIN, not by SYK. SYK populates and stabilizes the substrate.

8.6 Correction 6 - The role of SYK in the overall architecture

Previous statement. SYK TENSOR papers imply that SYK produces the four-dimensional vacuum.

Corrected statement. The $D = 4$ substrate is derived by FDLM/WIN from the photon's two helicity states. SYK does not produce the four-dimensional geometry. It provides the microscopic fermionic degrees of freedom, the antipode operator identity, the maximal chaos, and the JT dual.

9 Conclusion

We have computed the large- N infrared structure of the SYK array on the 8×8 torus. For the standard uncorrelated disorder ensemble, the melon self-energy is exactly position diagonal, and consequently momentum-independent. The theory flows to the 0+1-dimensional SYK conformal fixed point with $\Delta = 1/4$. The dual is AdS_2 . No four dimensional gravity emerges from the SYK array.

We have verified this conclusion numerically at $N = 16$ with momentum-conserving disorder. The Green's function is position-independent to 0.64%. The momentum-conserving constraint is too weak to preserve two-dimensional structure in the infrared.

We have stated the corrected architecture of the WIN Paradigm. FDLM/WIN provides the $D = 4$ substrate. SYK/QIN provides the microscopic population and stabilization mechanism. The two roles are distinct, and conflating them has been the source of the incorrect claims in previous papers.

We have catalogued six corrections: the value of γ , the Gaussian variance, the relation between γ and b , the chaos-gravity correspondence, the claim of 4D gravity from 2D SYK, and the role of SYK in the overall architecture. These corrections should be applied to previously circulated WIN papers.

The remaining open question in the SYK layer is the physical interpretation of the antipodal pairs. Whether the visible/hidden labeling of the 14 modes corresponds to a particle/antiparticle classification, to a confined/unconfined distinction, or to something else is not determined by the current analysis. We leave this for future work.

Acknowledgements

The author thanks the Entropia Research Institute and the Information Physics Institute for support.

A Numerical Values

Quantity	Value
Torus size	8×8
Sites	$N = 64$
Majorana fermions per site	1
Dimension	4
γ (DME)	$0.702839467990\dots$
γ (corrected UV)	0.62 ± 0.05
σ^2 (corrected)	$1/2$
Δ (IR scaling dimension)	$1/4$
Holographic dual	AdS_2 (JT gravity)
Numerical test N	16
$\mu_G(\beta/2)$	0.9531
$\sigma_G(\beta/2)$	0.0061
Fractional variation	0.64%

B Reproducible Verification

The following Python code reproduces the numerical result of Section 5.

```
import numpy as np
from scipy.linalg import eigh
from itertools import combinations

L = 4
N_sites = L * L
N_maj = 16
dim = 2 ** (N_maj // 2)

def majorana_ops(N):
    n = N // 2
    I = np.eye(2, dtype=complex)
    X = np.array([[0, 1], [1, 0]], dtype=complex)
    Y = np.array([[0, -1j], [1j, 0]], dtype=complex)
    Z = np.array([[1, 0], [0, -1]], dtype=complex)
    def kron_all(ops):
        r = ops[0]
        for o in ops[1:]:
            r = np.kron(r, o)
        return r
    g = []
    for j in range(n):
        zs = [Z] * j
        g.append(kron_all(zs + [X] + [I] * (n - j - 1)))
        g.append(kron_all(zs + [Y] + [I] * (n - j - 1)))
    return g
```

```

gammas = majorana_ops(N_maj)

def site_to_k(s):
    return (s // L, s % L)

def make_J_momentum_conserving(L, J=1.0, seed=42):
    rng = np.random.default_rng(seed)
    N = L * L
    sigma = np.sqrt(6 * J**2 / N**3)
    J_dict = {}
    for i, j, k, l in combinations(range(N), 4):
        ki, kj, kk, kl = (site_to_k(i), site_to_k(j),
                           site_to_k(k), site_to_k(l))
        tot = ((ki[0] + kj[0] + kk[0] + kl[0]) % L,
               (ki[1] + kj[1] + kk[1] + kl[1]) % L)
        if tot == (0, 0):
            J_dict[(i, j, k, l)] = rng.normal(0, sigma)
    return J_dict

Jdict = make_J_momentum_conserving(L)
H = np.zeros((dim, dim), dtype=complex)
for (i, j, k, l), J in Jdict.items():
    H += J * (gammas[i] @ gammas[j] @ gammas[k] @ gammas[l])

evals, evecs = eigh(H)
beta = 2.0
boltz = np.exp(-beta * (evals - evals.min()))
boltz /= boltz.sum()

def G_local(site, tau, evals, evecs, boltz):
    g = gammas[site]
    G = 0.0
    for n in range(len(evals)):
        for m in range(len(evals)):
            amp = np.abs(evecs[:, n].conj() @ (g @ evecs[:, m]))**2
            G += boltz[n] * amp * np.exp(tau * (evals[n] - evals[m]))
    return G

tau = beta / 2
G_vals = [G_local(s, tau, evals, evecs, boltz) for s in range(N_maj)]
print(f"G_local(tau=beta/2) per site:")
print(f"  mean = {np.mean(G_vals):.4f}")
print(f"  std  = {np.std(G_vals):.4f}")
print(f"  min  = {np.min(G_vals):.4f}")
print(f"  max  = {np.max(G_vals):.4f}")
print(f"  std/mean = {np.std(G_vals) / np.mean(G_vals):.6f}")

```

Output:

```

G_local(tau=beta/2) per site:
mean = 0.9531
std  = 0.0061
min  = 0.9424
max  = 0.9637
std/mean = 0.006382

```

C Summary of Corrections

Paper	Correction
SYK_TENSOR_TOE_II	$\gamma = 0.703$, not 1.0–1.2; remove $G_{\text{eff}} = 1/\lambda_L$; remove 4D gravity claim
SYK_TENSOR_TOE	Same
SYK_TENSOR_BLACK_HOLE	Same
SYK_TENSOR_FORCE_UNIFICATION	Change structure from FDLM/WIN, not SYK
cosmological_constant	Weaken $1/N^2$ claim; remove chaos-gravity mechanism
UV.pdf	$\sigma^2 = 1/2$; $\gamma = -b$
TOE_2_7_2026	$\sigma^2 = 1/2$; $\gamma = -b$
WIN_Paradigm_Explained	Add architecture note; update §11.3, §11.4
The_SYK_Antipode_Theorem	No change
The_SYK_Antipode_Correspondence	No change (superseded by theorem paper)

References

- [1] S. Sachdev and J. Ye, “Gapless spin-fluid ground state in a random quantum Heisenberg magnet,” *Phys. Rev. Lett.* **70**, 3339 (1993).
- [2] A. Kitaev, “A simple model of quantum holography,” KITP talks (2015).
- [3] J. Maldacena and D. Stanford, “Remarks on the Sachdev-Ye-Kitaev model,” *Phys. Rev. D* **94**, 106002 (2016).
- [4] K. Jensen, “Chaos in AdS_2 holography,” *Phys. Rev. Lett.* **117**, 111601 (2016).
- [5] J. Maldacena, S. H. Shenker, and D. Stanford, “A bound on chaos,” *JHEP* **08**, 106 (2016).
- [6] S. Preschutti, “The WIN Paradigm Explained: A Structural Overview,” WIN Paradigm White Paper (2026).
- [7] S. Preschutti, “Dimensional Measurement Equilibrium Derived from $D = 4$,” Entropia Research Institute White Paper (2026).
- [8] S. Preschutti, “ENTROPIA-Core: Dynamical Stability of the DME Vacuum,” Entropia Research Institute Preprint (2026).
- [9] S. Preschutti, “ENTROPIA-EFT: The Effective Field Theory of the DME Vacuum,” Entropia Research Institute White Paper (2026).
- [10] S. Preschutti, “ENTROPIX: A Master Entropy Selection Framework,” Entropia Research Institute White Paper (2026).
- [11] S. Preschutti, “The SYK-Antipode Theorem: An Exact Operator Identity for the 14 Torus Modes at $\lambda = 4$,” Entropia Research Institute Preprint (2026).
- [12] S. Preschutti, “ENTROPIX/MESA Rigorous UV Completion and Stability of Soft Regulated Entanglement Entropy in Non-Markovian Flows,” Entropia Research Institute (2026).
- [13] S. Preschutti, “ENTROPIX/MESA Soft Regulated Entanglement Entropy in Open Quantum Systems: A Framework for Bounded Variance and UV Control,” Entropia Research Institute (2026).

- [14] S. Preschutti, “The SYK TENSOR Quantum Information Network (QIN): Resolution of the Black Hole Information Paradox,” Entropia Research Institute (2026).
- [15] S. Preschutti, “The SYK TENSOR Quantum Information Network (QIN): Force Unification in a Chaotic Qubit Substrate,” Entropia Research Institute (2026).
- [16] S. Preschutti, “The SYK TENSOR Quantum Information Network (QIN): Resolution of the Cosmological Constant Problem and Dark Energy,” Entropia Research Institute (2026).